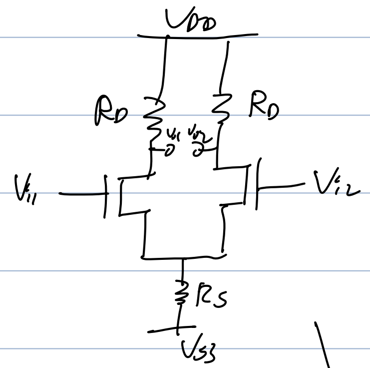
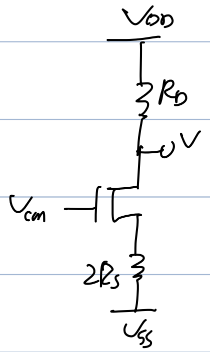
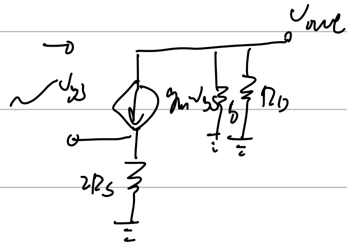
Diff Amp

Long tail pair diff amp with balanced I/O



CM





$v_{out} = -g_m v_{gs}(r_o // R_D)$

$v_{gs} = v_{cm} - g_m v_{gs} \cdot 2R_s$

$\Rightarrow v_{gs} = \dfrac{v_{cm}}{1 + g_m \cdot 2R_s}$

$\Rightarrow A_{cm} = \dfrac{v_{out}}{v_{cm}} = -\dfrac{g_m(r_o//R_D)}{1+2g_mR_s} \approx -\dfrac{R_D}{2R_s}$

diff

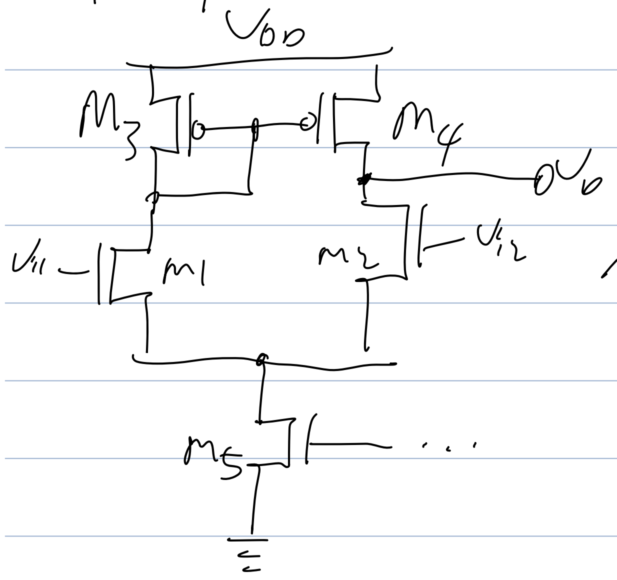
NO Rs virtual ground

$\Downarrow$

$A_{diff} = -\dfrac{g_m(r_o//R_D)}{1} \approx -g_m R_D$

$CMRR = \dfrac{A_{diff}}{A_{cm}} = g_m R_D \Big/ \dfrac{R_D}{2R_s} = 2 g_m R_s$

Diff amp with active load



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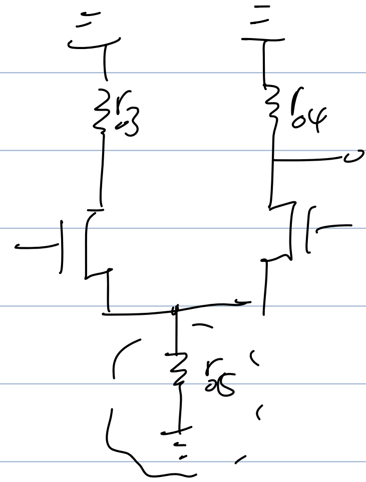

$$\Rightarrow V_{o,cm} \approx \frac{-r_{o4}}{2r_{o5}}$$

$$V_{o,diff} = \frac{1}{2} g_m (r_{o2} // r_{o4})$$

$$\Rightarrow |CMRR| \approx |\ \ | = g_m \cdot \frac{r_{o2} r_{o5}}{r_{o2} + r_{o4}}.$$